

CSCI 4266 - Final Project Report

An Exploration of Simplified Deep Learning Models for Protein Secondary Structure Prediction

Matt Schneider & Kevin White

December 2025

1 Introduction

1.1 Scientific Motivation

Proteins are the functional engines of biological systems, driving everything from cellular signaling and metabolism to structural organization. Their behavior, however, is not determined by sequence alone. Instead, it emerges through a complex folding process that maps a one-dimensional chain of amino acids into a stable three-dimensional structure. While experimental methods such as X-ray crystallography and NMR spectroscopy can resolve these structures, they remain time-consuming, expensive, and infeasible at the scale modern biology demands [18].

Protein secondary structure prediction offers a more accessible stepping stone: instead of generating full 3D geometries, models classify each residue into coarse structural categories such as helices, strands, and coils [5]. These classifications provide insight into protein function, stability, and interactions, and they are core components of larger pipelines that reason about full tertiary structures [1]. Despite being a simplified problem relative to full protein folding, secondary structure prediction is scientifically meaningful because it captures the local building blocks from which global structure emerges.

1.2 The Gap: Computational Constraints in Modern Protein Modeling

In recent years, the landscape of protein modeling has changed dramatically. Large-scale transformer architectures, such as AlphaFold2 and protein language models like ProtT5 [2], have set new performance standards by leveraging millions of parameters, massive datasets, and advanced evolutionary information. These models achieve near-experimental accuracy, but only under computational conditions far beyond what a typical research environment or classroom setting can replicate.

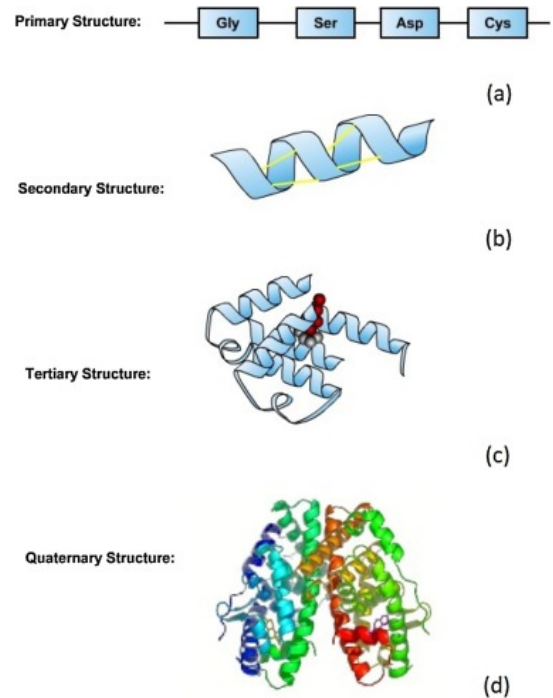


Figure 1: Overview of protein structure prediction process.

This creates a gap in the field between what is scientifically possible and what is computationally accessible. Students and smaller research groups often rely instead on traditional deep learning architectures like convolutional neural networks (CNNs), recurrent neural networks (RNNs), and classical evolutionary features like position-specific scoring matrices (PSSMs) to provide local patterns, sequence-level dependencies, and sequence specific mutation information. These models are less expressive, but they offer an important opportunity to study the underlying principles of protein structure prediction without relying on industrial-scale computational resources.

Our project resides within this gap. We attempt to build and evaluate a secondary structure prediction system using a combination of CNN and BiLSTM architectures. These are approaches that predate today’s large language models but remain influential in biological sequence modeling. By working under real computational constraints, we hope to better understand the strengths, limits, and challenges of classical architectures in the modern era of protein AI.

1.3 Objectives

The primary objective of this project is to implement, train, and evaluate a deep learning pipeline capable of predicting Q3 and Q8 secondary structure labels from raw amino acid sequences. Specifically, we aim to:

1. Construct baseline convolutional and bidirectional LSTM models for residue-level classification.
2. Combine these architectures into a hybrid CNN-BiLSTM model intended to capture both local motifs and global sequence context.
3. Incorporate evolutionary information through PSSMs to investigate whether classical sequence profiles still provide measurable benefits.
4. Evaluate performance on the PS4 dataset for training and the CB513 benchmark for standardized comparison.
5. Analyze the causes of model failure, instability, or underperformance. This includes any data issues, architectural limitations, and computational bottlenecks.

2 Data

2.1 PS4 & CB513 Data Sets

The previous state of the art dataset was the CB513 dataset [19]. This dataset consists of 513 secondary protein structures and their amino acid residues, determined with x-ray crystallography. This dataset is still often used as the academic standard for model validation in this field due to the high accuracy of the experimentally determined structures. We also used this dataset as a benchmark to validate our models performance.

As mentioned, this field has rapidly shifted towards the use of data-hungry transformer based models. To provide LLM based models with large amounts of data, Omar Peracha from Oxford University has created the largest dataset for protein secondary structure prediction: PS4 [1]. This dataset contains 18,731 nonredundant protein chains and their respective secondary structure labels. Each chain is identified in a protein database using a unique label. All entries within this dataset are nonredundant against other common secondary structure datasets. The advantage of the PS4 dataset, as described in the paper, is that models trained on it is able to achieve state of the art accuracy on both Q3 and Q8 prediction tasks, without the use of multiple sequence alignment (MSA) data. This was the particular reason we chose this dataset. We aimed to train our models on *only* amino acid sequences (primary structure) and the labeled secondary structure of such sequences. Similarly, to incorporate peer feedback, we aimed to find the largest dataset possible.

2.2 Position Specific Scoring Matrices

To incorporate more information into our models to accommodate their simple architecture, we aimed to include position specific scoring matrices for each amino acid chain. These matrices are built from a multiple sequence alignment, for each amino acid sequence in the PS4 dataset. We created this data by first searching the NCBI BLAST database using the protein database label for a given entry in the PS4 dataset. This allowed us to find homologous sequences, from which we aligned them together to create the MSA. From this multiple sequence alignment we are able to count the frequency of each amino acid (residue) at each position in the MSA. This gives us information on how common each amino acid is at a specific position in a given sequence. For this information to be useful, we need to decide on a background frequency model for all 20 amino acids. Creating this model is simple, we just count the number of occurrences for each amino acid in the PS4 dataset, and divide by the total number of amino acids.

In our secondary structure prediction models, we chose to incorporate PSSM features so that even relatively simple architectures could exploit rich evolutionary information. PSSMs summarize how conserved or variable each residue position is across homologous sequences, providing a compact representation of which amino acids are favored or disfavored at each site. This evolutionary signal has been shown to be highly informative for secondary structure, and we therefore follow prior work such as Asgari et al. [17] in using it to enhance our input features. By adding PSSMs, we allow the model to indirectly capture long-range and context-dependent effects that arise during protein folding; patterns that low-capacity models would be unlikely to learn from single sequences alone.

3 Methods & Approach

3.1 Overview

Our project sought to implement a classical deep learning pipeline for residue-level secondary structure prediction using only raw amino acid sequences and, when possible, evolutionary features. The approach deliberately avoided large transformer-based pretrained models, both to reduce computational demand and to evaluate the capabilities and limitations of smaller architectures that have historically been used in protein modeling. To this end, we designed, trained, and evaluated three models: a convolutional neural network (CNN), a bi-directional long short-term memory network (BiLSTM), and a hybrid CNN-BiLSTM model that stacked convolutional feature extractors with recurrent sequence processing layers.

Each model was implemented in PyTorch and trained on the PS4 dataset, with evaluation performed on the CB513 benchmark. All training was conducted on local hardware with limited GPU memory, which imposed practical constraints on batch size, embedding dimensions, layer depth, and the incorporation of evolutionary features such as position-specific scoring matrices (PSSMs).

3.2 Data Representation and Encoding

For each protein chain, the input sequence was represented as a string of single-letter amino acid codes. These characters were mapped to integer token IDs and passed through a learnable embedding layer, yielding dense vectors that served as the model’s per-residue input features. Embeddings allowed the model to position amino acids within a continuous feature space in which residues with similar biochemical properties could be represented more closely.

To generate position-specific features for the PS4 dataset, we implemented the pipeline described in Section 2.2 using the NCBI BLAST web API. For each protein sequence in PS4, we submitted a BLASTP query against a protein database and collected the homologous sequences returned by BLAST. From these alignments, we constructed a position-specific scoring matrix (PSSM) by counting, for each position in the query sequence and each amino acid type, how often that amino acid appeared in the corresponding alignment column. We then converted these counts into log-odds scores relative to a background amino acid distribution, yielding an $L \times 20$ matrix for a sequence of length L , where rows correspond to sequence positions and columns correspond to the 20 standard amino acids.

Because the NCBI BLAST service is not intended for large-scale batch processing and is subject to rate limits (see

Section 5), we were only able to generate PSSMs for a small subset of the 18,731 PS4 sequences using the online API. Nevertheless, these examples are sufficient to illustrate the type of information captured by the PSSM. As shown in Figure 2, some positions exhibit a strong preference for a single amino acid, with one column showing high positive scores and most alternatives showing strongly negative scores; such positions are likely under strong evolutionary constraint and are highly conserved among homologous sequences. Other positions exhibit a flatter distribution of scores across multiple amino acids, indicating that substitutions at those sites are more tolerated. In the particular example visualised, positions containing tryptophan (W) tend to show strong conservation, whereas some positions containing leucine (L) appear more permissive to alternative residues. Importantly, these observations are *position-specific* within the given protein, and should not be interpreted as global claims about the mutability of W or L across all proteins.

These position-specific preferences are valuable to our models because they provide information about which regions of the sequence are under strong evolutionary constraint and which regions are more flexible. Substitutions at highly conserved positions are more likely to disrupt the native fold or function, while substitutions at more tolerant positions may have milder structural effects. By incorporating PSSM profiles as input features, the model can exploit these evolutionary signals when predicting secondary structure. We note, however, that our profiles are derived from a single BLASTP search (rather than a full iterative PSI-BLAST procedure) and use a simplified uniform background model; thus, they should be viewed as approximate position-specific substitution preferences rather than canonical PSI-BLAST PSSMs.

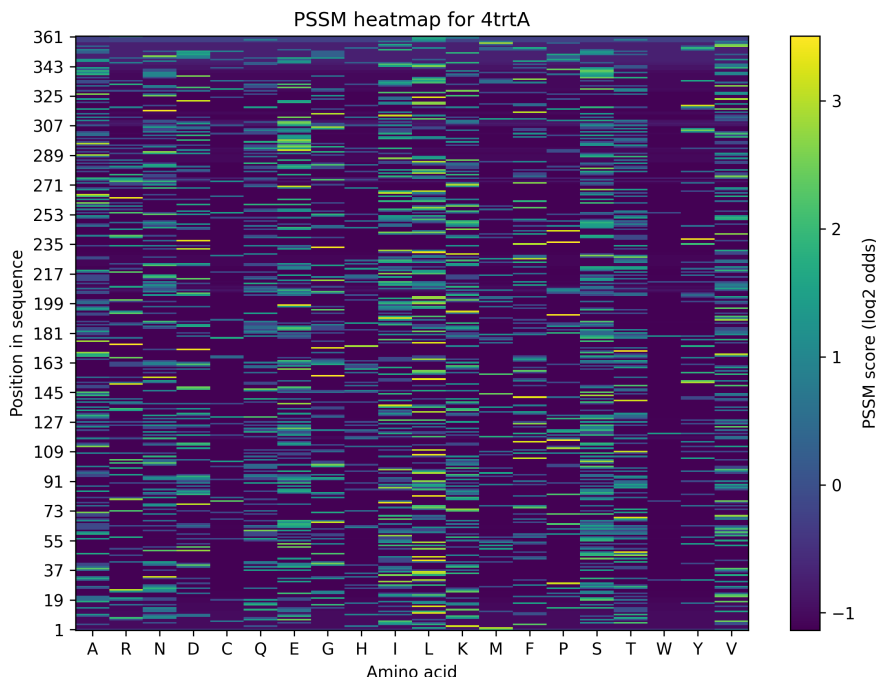


Figure 2: PSSM for sequence id: 4trtA of PS4 dataset

3.3 Evaluation Metrics: Q3 & Q8 Accuracy

Prediction models use two main validation metrics, Q3 and Q8 accuracy. These metrics describe how many structure classes are in the final prediction. Q3 accuracy splits secondary structure into three main classes of protein residues: Helix (**H**), Strand/Beta-sheet (**E**), and Coil/Other (**C**). Similarly, Q8 refines these classes into more nuanced detail and is a more difficult prediction task. These classes relate to the final shape of the amino acids within their predicted structure. Specifically, these are the eight classes used in the Q8 prediction task: **H** α -helix, **G** 3_{10} -helix, **I** π -helix, **E** extended β -strand, **B** detached β -strand/ β -bridge, **T** turn, **S** bend, **C** coil/other (irregular structures) [5] [8].

Ground-truth labels for each residue were provided in the DSSP8 format, an eight-state secondary structure system (Q8) that distinguishes fine-grained helix, strand, and coil subtypes. For certain analyses, these labels were also

collapsed into the more common three-class DSSP3 scheme (Q3), which merges them into helix (H), strand (E), and coil (C). Mask tokens were used to ensure that padded positions in batches did not contribute to training loss.

3.4 CNN Model: Local Pattern Extraction

The first model implemented was a 1D convolutional neural network designed to capture local spatial patterns within amino acid sequences. CNNs have historically been effective at identifying short-range motifs such as turns, helix nucleation sites, and strand β -pairing signals.

Our implementation used multiple convolutional kernels of varying widths (e.g., 3, 5, 7, and 11) to let the network detect motifs across different contextual windows. Each kernel produced a separate set of feature maps, and these were concatenated along the channel dimension. Padding ensured that the output length matched the input sequence length so that residue-level alignment was preserved. After convolution, dropout and a linear projection layer were applied to stabilize training and reduce overfitting.

The CNN served as both a baseline model and as the feature extractor for the hybrid architecture described later. Its performance reflected the expected behavior of convolutional models: strong ability to capture local structure but limited capacity for modeling long-range dependencies.

3.5 BiLSTM Model: Sequential Dependency Modeling

The second model was a bidirectional LSTM (BiLSTM), chosen for its ability to incorporate sequential context from both the N-terminal and C-terminal directions. A single LSTM pass processes residues from left to right, generating a hidden representation that encodes information from the prefix of the sequence. A bidirectional LSTM augments this with a reverse pass, producing a second set of hidden states informed by downstream residues. The concatenation of forward and backward states enables the model to make predictions using both past and future context, a property that is important for residue-level classification tasks.

The BiLSTM architecture consisted of two recurrent layers with dropout applied between them. While these networks are theoretically capable of modeling long-range interactions relevant to folding, their expressiveness is limited by hidden-state dimensionality and the vanishing gradient problem. In practice, GPU memory constraints forced us to adopt smaller hidden sizes and batch dimensions than would be ideal, which likely constrained overall performance.

3.6 Combined CNN-BiLSTM Pipeline

Motivated by literature on secondary structure prediction, we implemented a hybrid model that stacked the CNN feature extractor directly before the BiLSTM. The intuition behind this design is that convolutional layers efficiently extract local structural cues, while the BiLSTM integrates these local features into a broader contextual representation that approximates global folding patterns.

The hybrid model begins with the same multi-kernel CNN, producing a concatenated feature map. This representation is fed into the BiLSTM, allowing sequential processing of the enriched local features. A final linear layer maps the resulting hidden states to an 8-dimensional output corresponding to the Q8 classes. Softmax was applied to produce per-residue class probabilities.

3.7 Training Details

All models were trained using cross-entropy loss with class weighting, where weights were computed as the inverse frequency of each DSSP8 label in the training set. This adjustment compensates for the severe class imbalance—particularly the under-representation of β -bridges and π -helices, and prevents the model from collapsing into predicting only the most common states. We used the Adam optimizer with a learning rate on the order of 10^{-3} ,

batch sizes ranging from 32 to 128 (depending on GPU memory availability), and early stopping based on validation Q8 accuracy.

Training was conducted on an NVIDIA RTX 5080 GPU, but memory limitations frequently restricted the model size and prevented extensive hyperparameter sweeps. The inability to train deeper models or stack additional BiLSTM layers likely prevented the architectures from reaching their full potential. Additionally, sequences were padded to the maximum length in each batch; while masking prevented padded tokens from influencing gradients, the varying sequence lengths introduced overhead that reduced computational efficiency.

4 Results

Our study evaluated three architectures for 8-class (Q8) and 3-class (Q3) protein secondary structure prediction: a 1D convolutional neural network (CNN), a bidirectional LSTM (BiLSTM), and a hybrid ensemble combining both models. Performance was assessed first on the PS4 validation split used during training, and then on the CB513 benchmark dataset to measure real-world generalization. Figures 5b and 5a summarize these outcomes.

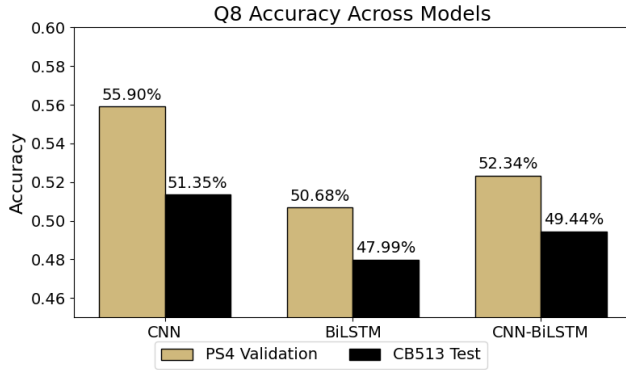


Figure 3: Q8 Accuracy performance of each model on PS4 validation vs CB513 test.

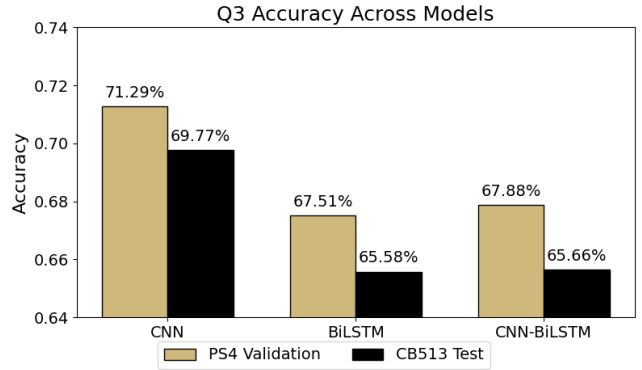


Figure 4: Q3 Accuracy performance of each model on PS4 validation vs CB513 test.

4.1 Validation Performance on PS4

Across all metrics, the CNN achieved the strongest validation performance. For Q8 classification, the CNN reached 55.90%, outperforming the BiLSTM (50.68%) and the hybrid (52.34%). A similar pattern appeared in Q3 accuracy, where the CNN reached 71.29%, compared to 67.51% for the BiLSTM and 67.88% for the combined model. These results suggest that the convolutional architecture was more effective at capturing residue-local motifs in the PS4 dataset than the recurrent alternative.

4.2 Generalization to CB513

When tested on the CB513 dataset, all models exhibited a drop in accuracy, indicating limited generalization. The CNN again performed best, with 51.35% Q8 accuracy and 69.77% Q3 accuracy. The BiLSTM and hybrid ensemble performed worse, achieving 47.99% / 65.58% (Q8/Q3) and 49.44% / 65.66% respectively. This degradation aligns with known challenges in secondary structure prediction: models that rely heavily on residue-local patterns often fail to generalize across datasets with different fold distributions or sequence homologies.

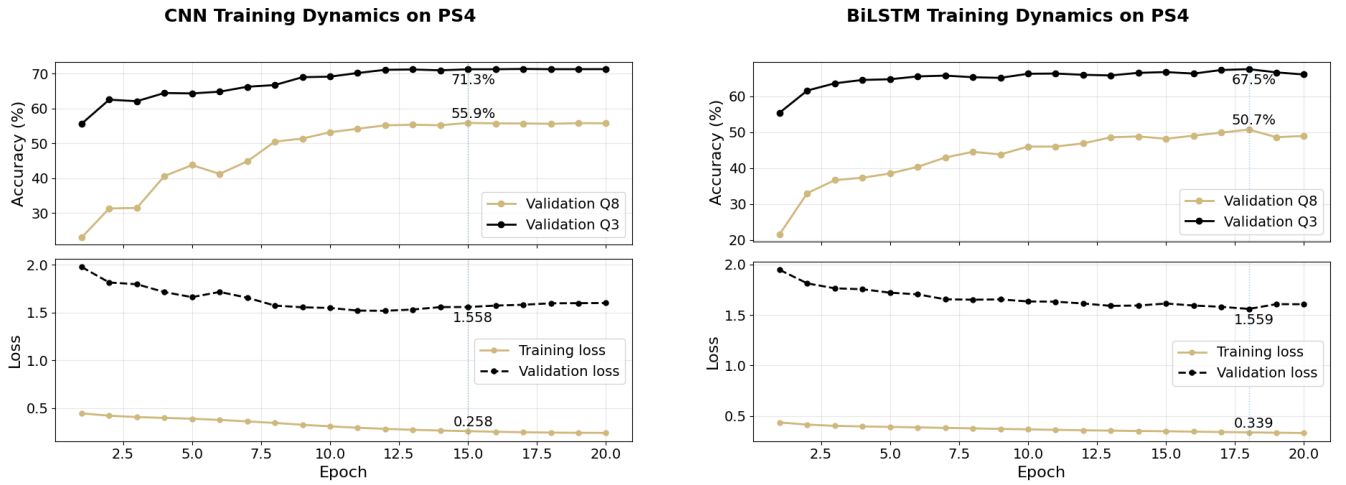
Notably, the hybrid model did not improve over the CNN alone. While combining convolutional and recurrent features is a well-motivated strategy, our implementation did not yield complementary strengths. Instead, it appears that the weaknesses of the BiLSTM component, in particular its difficulty modeling long-range structure without evolutionary features, reduced the ensemble’s effectiveness relative to the CNN.

4.3 Interpretation and Comparison Across Models

Taken together, the results show that: (1) The CNN consistently performed best across both validation and test datasets. (2) The BiLSTM underperformed, likely due to insufficient contextual information in raw amino acid sequences. (3) The hybrid model did not meaningfully improve over either base model, suggesting that simply concatenating learned representations is not enough to leverage potential complementarities. (4) All three models generalized poorly to CB513, indicating that the PS4 dataset alone does not provide enough sequence diversity or evolutionary context for strong out-of-distribution performance.

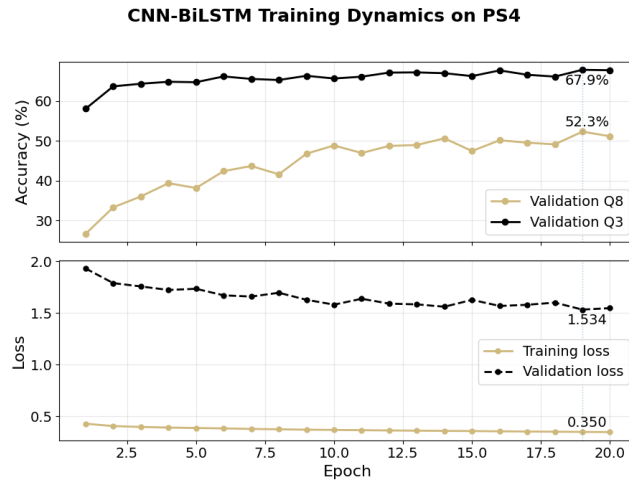
These findings align with prior work showing that models trained solely on one-hot amino acid input struggle to bridge the gap between training datasets and standardized benchmarks unless supplemented with richer biologically informed features such as PSSMs or transformer-based embeddings.

4.4 Training Dynamics



(a) CNN model accuracy and loss over training epochs.

(b) BiLSTM model accuracy and loss over training epochs.



(c) CNN-BiLSTM model accuracy and loss over training epochs.

Figure 5: Model accuracy and loss over training epochs on the PS4 dataset.

Across all three architectures, the training curves revealed consistent patterns that help explain the models’ final performance. For the CNN, training and validation accuracy increased smoothly before plateauing at approximately 56% Q8 and 71% Q3, with the corresponding loss curves flattening after several epochs. Notably, the gap between training and validation accuracy remained relatively small, suggesting that the CNN learned stable residue-level patterns without heavily overfitting to the PS4 dataset.

The BiLSTM displayed a different trajectory: both training and validation accuracy rose more slowly, and the model converged to lower overall accuracy. The training curves suggest that the BiLSTM had difficulty optimizing purely from sequence input, likely because long-range structural dependencies cannot be captured effectively without evolutionary profiles or transformer-style contextual embeddings. The hybrid CNN-BiLSTM model followed an intermediate pattern, improving more slowly than the CNN but plateauing similarly, ultimately failing to surpass either component model.

Together, the training plots indicate that none of the models overfit heavily, but they also did not discover increasingly useful representations over time, plateauing early and suggesting that raw amino acid sequences alone provide insufficient signal for deeper learning. These learning dynamics support the later conclusion that feature engineering, in particular the use of PSSMs or pretrained protein language models, would likely be necessary for improved generalization.

Across all three models, training loss decreased steadily over the 20-epoch schedule, indicating that each architecture successfully fit the PS4 training data. However, validation loss flattened early and remained substantially higher than training loss, never dropping below 1.55 for any model, which signals slight overfitting. The combined CNN-BiLSTM model showed slightly lower validation loss than the standalone CNN or BiLSTM, but the gap between training and validation loss persisted, supporting that none of the architectures were able to generalize well to unseen sequences despite continued improvement on the training set.

4.5 Conclusion

Our findings show a contradictory result to the papers we studied, specifically [17], [1], [13], and [7]. Despite its simplicity, our CNN achieved the best performance on PS4 when using only primary sequence to predict secondary structure, outperforming both the BiLSTM and the hybrid CNN-BiLSTM models. This result contrasts with several papers we reviewed, where recurrent or hybrid architectures consistently outperformed plain convolutional models on similar tasks. Our findings suggest that, in the low-feature regime (no PSSMs or pretrained embeddings) and with a dataset of this size, a strong local inductive bias and lower model capacity can be more advantageous than more expressive sequence models. This highlights the importance of matching model complexity and architecture to both the available data and the information content of the input features, rather than assuming that more complex models will always perform better. Unfortunately, our models displayed clear signs of overfitting to training data and had significant performance drops when applied to the unseen CB513 dataset.

5 Discussion

As learned through our extensive research into this topic, the complex dynamics of the protein folding problem requires models that are able to accurately and precisely capture complex chemical relationships. Capturing these relationships requires large scale models with massive model sizes and many many parameters. These requirements enforce a large baseline compute capability, to both run and train these models. During the course of this project, we faced many challenges with our limited compute resources and ability to replicate the state of the art model architectures.

Our first challenge came to the model complexity we planned to develop. As briefly mentioned in section 3.7, we had to reduce the model complexity in order to accommodate the power of our single NVIDIA RTX 5080 graphics card. This likely caused us to lose performance when comparing to the models of similar architecture trained in

The most critical limitation to our study was the generation of our PSSM data based on the massive PS4 dataset. Although this dataset contains plenty of secondary structure information for each sequence, the simple architecture

of our model required additional features like PSSMs. As described in section 2.2 and, the process to create these features for each sequence did not go according to plan. We faced major challenges generating these matrices for each entry in our dataset due to the complex and computationally expensive process required to create each PSSM. There were two methods we examined to generate the PSSMs, the method described in section 2.2, and an alternate method that exchanges computation speed for storage capacity. The alternate method required installing NCBI BLAST+, a local software package that contains the same API features as before, and additional features for creating PSSMs and other bioinformatics tools. In order to use these additional tools, we attempted to create a local extended version of the swissprot [21] database used from creating our PSSMs in the API method. This proved to be just as difficult and expensive as waiting for API requests from NCBI BLAST. The storage space required to create this database was approximately 800GB, and would only lower the compute time for generating the PSSMs to a conservative estimate of 10 days, instead of a conservative estimate of 15. For these reasons, our models did not include PSSM features as we originally planned.

6 References

- [1] Peracha, Omar. “PS4: A Next-Generation Dataset for Protein Single-Sequence Secondary Structure Prediction.” *BioTechniques*, vol. 76, no. 2, 2024, pp. 63–70. Taylor & Francis Online, doi:10.2144/btn-2023-0024.
- [2] Elnaggar, Ahmed, et al. “ProtTrans: Toward Understanding the Language of Life through Self-Supervised Learning.” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 10, 2022, pp. 7112–7127. IEEE Xplore, doi:10.1109/TPAMI.2021.3095381.
- [3] AlQuraishi, Mohammed. “ProteinNet: A Standardized Data Set for Machine Learning of Protein Structure.” *BMC Bioinformatics*, vol. 20, 2019, article 311. *ProteinNet* GitHub Repository, <https://github.com/aqlaboratory/proteinnet>.
- [4] Torrisi, Mirko, Gianluca Pollastri, and Qian Kong. “Deeper Profiles and Cascaded Recurrent and Convolutional Neural Networks for State-of-the-Art Protein Secondary Structure Prediction.” *Scientific Reports*, vol. 9, 2019, article 12374, <https://www.nature.com/articles/s41598-019-48786-x>.
- [5] Sun, Peter D., Christine E. Foster, and Jeffrey C. Boyington. “Overview of Protein Structural and Functional Folds.” *Current Protocols in Protein Science*, vol. 35, no. 1, 2004, unit 17.1. Wiley, doi:10.1002/0471140864.ps1701s35.
- [6] “Position Weight Matrix.” *Wikipedia*, Wikimedia Foundation, https://en.wikipedia.org/wiki/Position_weight_matrix. Accessed 11 Dec. 2025.
- [7] Wang, Lei, et al. “A Comprehensive Review of Protein Language Models.” *arXiv*, 8 Feb. 2025, arXiv:2502.06881, doi:10.48550/arXiv.2502.06881.
- [8] “Lecture11 – Protein Language Models – MLCB24.” *YouTube*, <https://www.youtube.com/watch?v=uPoFdCUqBWk>. Accessed 11 Dec. 2025.
- [9] Alanazi, Wafa, Di Meng, and Gianluca Pollastri. “DeepPredict: A State-of-the-Art Web Server for Protein Secondary Structure and Relative Solvent Accessibility Prediction.” *Frontiers in Bioinformatics*, vol. 5, 2025, article 1607402, doi:10.3389/fbinf.2025.1607402.
- [10] Weissenow, Konstantin, and Burkhard Rost. “Are Protein Language Models the New Universal Key?” *Current Opinion in Structural Biology*, vol. 91, 2025, article 102997, doi:10.1016/j.sbi.2025.102997.
- [11] Chen, Jia-Ying, et al. “Evaluating the Advancements in Protein Language Models for Encoding Strategies in Protein Function Prediction: A Comprehensive Review.” *Frontiers in Bioengineering and Biotechnology*, vol. 13, 2025, article 1506508, doi:10.3389/fbioe.2025.1506508.
- [12] “Autoregressive Model.” *Wikipedia*, Wikimedia Foundation, https://en.wikipedia.org/wiki/Autoregressive_model. Accessed 11 Dec. 2025.
- [13] Geffen, Yaron, Yanay Ofran, and Ron Unger. “DistilProtBert: A Distilled Protein Language Model Used to Distinguish between Real Proteins and Their Randomly Shuffled Counterparts.” *Bioinformatics*, vol. 38, suppl. 2, 2022, pp. ii95–ii98, doi:10.1093/bioinformatics/btac474.
- [14] Ouyang-Zhang, Jeffrey, et al. “Distilling Structural Representations into Protein Sequence Models.” *bioRxiv*, 2024, doi:10.1101/2024.11.08.622579.
- [15] Wang, Shuang, Jian Peng, and Jinbo Xu. “Protein Secondary Structure Prediction Using Deep Convolutional Neural Fields.” *Scientific Reports*, vol. 6, 2016, article 18962. PMC, <https://pmc.ncbi.nlm.nih.gov/articles/PMC4707437/>.
- [16] Ma, Kaixuan, et al. “Mega: Moving Average Equipped Gated Attention.” *arXiv*, 2022, arXiv:2209.10655, <https://arxiv.org/abs/2209.10655>.
- [17] Asgari, Ehsaneddin, et al. “DeepPrime2Sec: Deep Learning for Protein Secondary Structure Prediction from the Primary Sequences.” *bioRxiv*, 18 July 2019, doi:10.1101/705426.
- [18] Meiler, Jens, and David Baker. “Rapid Protein Fold Determination Using Unassigned NMR Data.” *Proceedings of the National Academy of Sciences*, vol. 100, no. 26, 2003, pp. 15404–15409, doi:10.1073/pnas.2434121100.
- [19] Cuff, J. A., and G. J. Barton. “Evaluation and Improvement of Multiple Sequence Methods for Protein Secondary Structure Prediction.” *Proteins: Structure, Function, and Genetics*, vol. 34, no. 4, 1999, pp. 508–519, <https://pubmed.ncbi.nlm.nih.gov/10081963/>.
- [20] National Center for Biotechnology Information (NCBI). “APIs.” *NCBI*, U.S. National Library of Medicine, <https://www.ncbi.nlm.nih.gov/home/develop/api/>. Accessed 11 Dec. 2025.
- [21] Salser, William, and Edgar T. Bolton. “Progress in DNA Research.” *Annual Review of Biochemistry*, vol. 41, 1972, pp. 173–214. PMC, <https://pmc.ncbi.nlm.nih.gov/articles/PMC102476/>.